Vista. EFS, however, is not a volume protector; it works on a file level. You cannot therefore use EFS to effectively protect sensitive areas of the system such as the Registry hive files.

In order to introduce encryption to the entire boot volume (including all files, Registry and otherwise, and data), Vista introduces Windows BitLocker Drive Encryption. Where EFS was implemented by the NTFS file system (as we mentioned, it worked on a file level), BitLocker encrypts at a volume level using a new driver called the Full Volume Encryption (FVE) driver. FVE sits between the NTFS file system and the data in the file system layer that it interacts with. It essentially acts as a filter, in that it transparently passes file requests from the NTFS layer to the layer below, and does its job on the fly: FVE sees all the I/O requests that NTFS sends to the volume, encrypting blocks as they're written and decrypting them as they're read. NTFS need not be even aware of the BitLocker protection since the encryption and decryption happen beneath NTFS and the encrypted the volume appears unencrypted to NTFS. If you attempt to read data from the volume from outside of Windows, however, it appears to be random data.

By default, volumes are encrypted using a 128-bit AES key called the Full Volume Encryption Key (FVEK). The FVEK in turn, is encrypted with a Volume Master Key (VMK). The VMK specifically can be protected in a number of ways, depending upon your system's hardware. A system with TPM (Trusted Platform Module) v1.2 support at the BIOS level can either encrypt the VMK with the TPM, have the system encrypt the VMK using a key stored in the TPM and one stored on a USB Flash device, or encrypt the key using a TPM-stored key and a PIN you enter when the system boots. For systems without TPM, you can encrypt the VMK using a key stored on an external USB Flash device.

Under BitLocker, the system goes through a verification chain, where each component in the boot sequence describes the next component to the TPM. Only if the entire chain of authorisation succeeds will the TPM allow access to the system. BitLocker there-